Sazzadul Islam

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Education

University of Chittagong June 2025 - Present
M.Sc. in Physics *Current GPA:*

University of Chittagong
B.Sc. in Physics *Current GPA: 3.34/4.0*

Coursework

Courses: Statistical Mechanics & Radiation, Computer Architecture and Programming, Quantum Mechanics I & II, Solid State Physics I & II, Computational Physics, Condensed Matter Physics, and Superconductivity.

Skills

Programming Languages: C++, Python, JavaScript, HTML/CSS

Simulation frameworks: Qiskit, PennyLane

Libraries: NumPy, Matplotlib

Formatting Application: LaTeX

Tools : GitHub, VS Code, draw.io, Origin

Research Interests

- Quantum Communication
- Quantum Information
- Quantum Simulation
- Quantum Teleportation

Research Internship and Workshops

ICTP Physics Without Frontiers (PWF) School on Quantum Field Theory (QFT) | *Participant* 2024
It was a month-long intensive program held online from August to October, 2024. The program featured lectures from international experts covering fundamental QFT, path integrals, and modern research applications, organized in collaboration with ICTP Physics Without Frontiers.

Glimpses into the Quantum World- Information, Technology, and Application | *Participant* 2024
IUB (Independent University Bangladesh) hosted a 2-week summer school split into pre-school and main school. Renowned physicists and leading experts delivered lectures on various tentative topics including Quantum Mechanics Essentials, Open Quantum Systems and Topological Phases of Matter.

Mahdy Research Academy (MRA) 2024
This year-long research course covered topics from basic to intermediate Quantum Information, Quantum Computing, and Quantum Machine Learning, along with hands-on experience on Qiskit coding.

Honors & Awards

- *WISER Summer Program 2026: WHAT IS THE FUTURE OF OPTIMIZATION? Classical + AI + Quantum-* - Awarded for completing 35+ hours of lectures, lab tours and industry projects and passing 5 modules.
- *IBM Quantum Excellence Badge (QGSS 2025)* — Awarded for completing all four core labs of the IBM Quantum Global Summer School 2025.
- *WISER Global Quantum Badge 2025* - Awarded for completing 35+ hours of lectures, lab tours and industry projects and passing 5 modules.

- *Womanium and WISER Quantum Program Challenges (2025)* — Completed PennyLane *Introduction to LCUs* and *A Simple Trotterization*.
- *Basics of Quantum Information* - Awarded for completing the foundational curriculum covering single- and multi-qubit systems, quantum circuits, and essential entanglement protocols like teleportation and superdense coding.

Volunteering & Extracurricular Activities

Bangladesh Physics Olympiad (BdPhO) | Script Evaluator (Category C) **2026**
Collaborated with the national committee to assess and score answer scripts for the national physics talent search. Maintained rigorous standards and accuracy when evaluating competitive exam papers.

Chittagong University Scientific Society (CUSS) | Core Committee Member **2023 – 2025**
Co-organized and executed 3 national-level Science Carnivals alongside multiple science outreach programs targeting a community of thousands of science enthusiasts across Bangladesh.